

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background.

A COMPREHENSIVE COVERAGE

MODULE - 18

AIR POLLUTION - 5

4 NITROGEN OXIDES (NO_x) AND NITROUS OXIDE (N_2O)

Nitric Oxide (NO) and Nitrogen Di Oxide (NO_2) together are known as NO_x (Nitrogen Oxides). Nitrous Oxide (N_2O) is also known as Laughing Gas.

SOURCES

- NO_x are produced from the reaction of oxygen and nitrogen at high temperatures ex: in the engines of automobiles.
- They are produced mainly in **internal combustion engines** and **coal burning power plants**. They are also produced naturally by **lightning**.
- Oxygen and nitrogen do not react at ambient temperatures. But at high temperatures they react and produce various oxides of nitrogen. Such temperatures arise inside an internal combustion engine or during lightning.



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SOURCES

- Burning of fossil fuels in automobiles, coal based thermal power plants, burning of agricultural waste, use of fertilizers which promotes nitrogen fixation by soil micro-organisms.
- Burning of crop residue contributes over 240 million kg of NO_x and 7 million kg of N_2O .
- N_2O is produced both naturally and through man-made activities.
- **Important natural sources are:** Soils under vegetation and the oceans.
- Natural sources contribute to 62% of the total emissions.
- Man-made sources contribute 38%.
- **Man-made sources of N_2O are:** agriculture, fossil fuels combustion, industries, burning of bio-mass etc.

The largest man-made source of N_2O emissions is Agriculture.

HOW FROM AGRICULTURE?

- The use of synthetic fertilizers in agriculture is a major source of N_2O emissions.
- When fertilizers are applied to the soil, the microbial processes of Nitrification and De-nitrification (Which are stages in Nitrogen cycle) produce N_2O , which is released into the atmosphere.
- Similarly, the use of manure as a fertilizer also leads to emissions from agricultural soils.

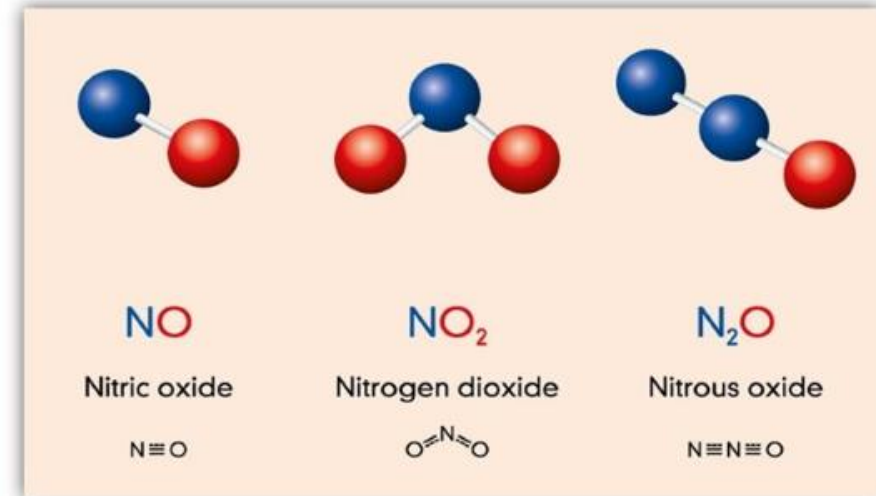


EFFECTS ON HEALTH

- NO_x aggravates asthmatic condition and create many respiratory health issues, especially in children.
- NO_x gases react with atmospheric gases to form smog and acid rain (Nitric acid) and also they are central to the formation of tropospheric ozone.

NO_x are not greenhouse gases, they do not cause global warming.

- Nitrous Oxide(N₂O) also known as Laughing Gas is a greenhouse gas and also severely impacts Ozone Layer.
- Nitrous Oxide is a greenhouse gas which causes global warming, it also causes ozone depletion.



5 SULPHUR DI OXIDE (SO₂)

It is a toxic gas with a pungent, irritating smell.

- It is released naturally by volcanic activity.
- It is abundantly available in the atmosphere of Venus.
- Burning of coal in thermal power plants is a major contributor to SO₂ emissions. Burning of diesel also causes emission of SO₂.
- Some industrial processes, such as production of paper and smelting of metals.
- The roasting of sulfide ores to such as pyrite, sphalerite, and cinnabar to extract pure sulphur.



Smelting is extracting a metal from its ore by a process involving heating and melting.

EFFECTS ON HEALTH

- Sulfur dioxide irritates the skin and mucous membranes of nose, throat, and lungs. It also irritates eyes.
- The resulting symptoms can include pain when taking a deep breath, coughing, throat irritation, and breathing difficulties.
- High concentrations of SO_2 can affect lung function, worsen asthma attacks, and worsen existing heart disease in sensitive groups.

EFFECTS ON ENVIRONMENT

- Sulphur Di Oxide is a powerful greenhouse gas, it causes global warming.
- SO_2 reacts with water vapour in the atmosphere to form sulphuric acid which falls to the ground as acid rain. It is the main cause of 'Marble Cancer' the Taj Mahal is currently facing.

MARBLE CANCER

- Marble cancer refers to the corrosion of buildings and statues made up of marble by action of sulphuric acid and nitric acid.
- It is caused by acid rain. Acid Rain results in corrosion by dissolving the calcium present in marble.
- Marble cancer causes disintegration and yellowing of the building or statue.



Taj Mahal is losing its white shine because of marble cancer.

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